How AI Effects the US Job Market

Aug Wu, Bouba Katompa, Bridgid Cutright

Motivation

- AI is rapidly changing how work is done in the US
- Some careers are threatened while others are created
- **Key Question:** Does higher AI exposure help or hurt careers?

Data Sources

- BLS Occupational Employment Data
  - Employment levels
  - Wage (25th, median, 75th)
- AI Exposure Score
  - Centered around 0
- Time period: 2022-2024

- EDA (clean and merge data)
- Calculate descriptive statistics
- Statistical testing

TOT_EMP_YOY_22	TOT_EMP_YOY_23	TOT_EMP_YOY_24
0.039920	0.022140	0.008313
0.054092	0.015245	0.011801
0.043256	0.029198	0.020425
0.045391	0.022309	0.013391

Research Questions

#1

Do high-AI-exposure jobs grow faster in employment?

#2

Do high-AI-exposure jobs hold up better on wages?

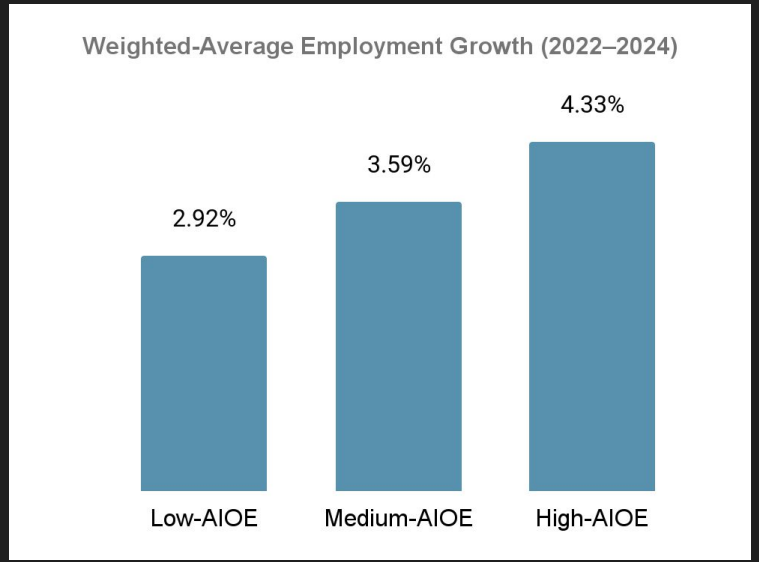
#3

Which are the top-performing occupations, what do they look like?

RQ1: High-AIOE Occupations Show Faster Employment Growth

Aggregated Occupation Tiers

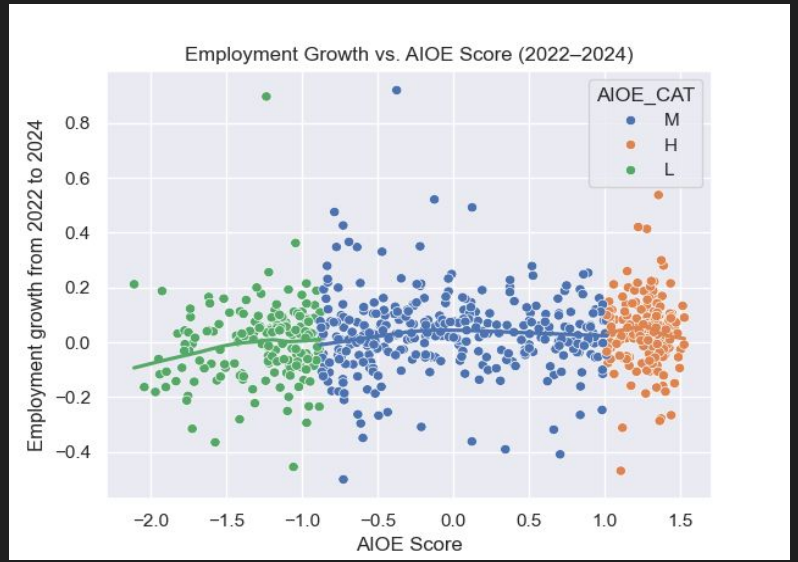
(weighted-average)



p-val = 6.03% (significant at 10%)

Individual Occupations

(unweighted-average)



p-val = 0.13% (significant at 10%)

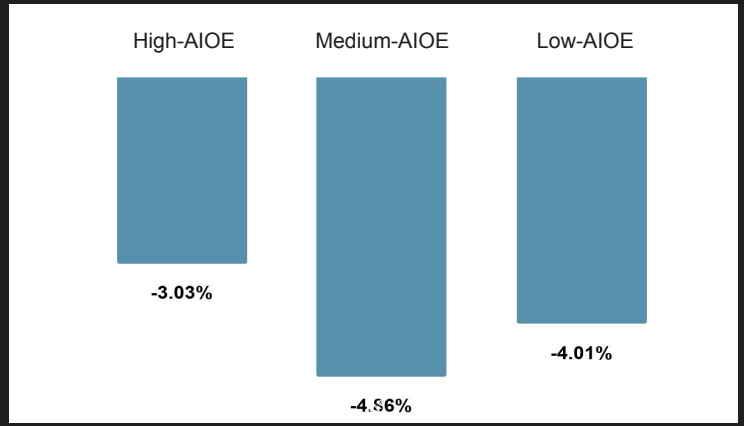
RQ2: High-AIOE Occupations Show Greater Wage Resilience

Wages rose & dispersion shrank across all AIOE tiers

We use wage dispersion growth to measure occupations' income competitiveness

Aggregated Occupation Tiers

(weighted-average)



p-val = 4.8% (significant at 5%)

Individual Occupations

(unweighted-average)

p-val = 1.6% (significant at 5%)

Wage Dispersion = (P75 – P25) / Median
Dispersion Growth = (Dispersion_{2024} – Dispersion_{2022}) / Dispersion_{2022}

RQ3 Result : which job perform best

Key Findings

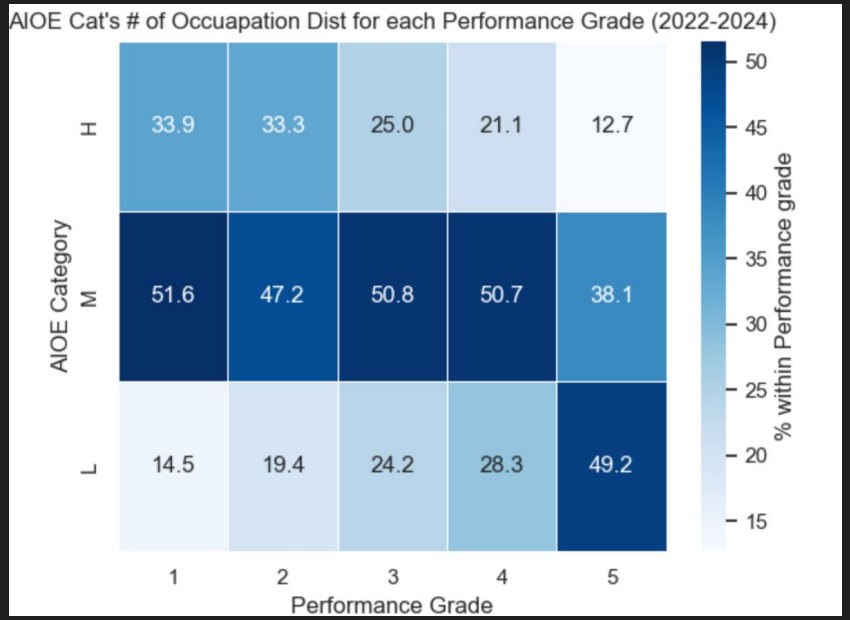
- Occupations were scored **1–5 using employment growth + wage dispersion growth**
- **High-AIOE jobs make up 33.9% of "very good" occupations** while representing only **25% of all jobs**
- Many top performers are **not tech roles**
  Examples: **Sales Managers, Civil Engineers, Medical & Health Services Managers**

Takeaway:

- You don't need to work **in AI** to benefit **from AI**

Heatmap Chart

(rq3_cnt)



Limitations + Future Work

Limitations

- Our analysis cannot fully separate three overlapping forces:
  AI adoption, post-COVID economic recovery, and broader U.S. economic restructuring

- These factors may all contribute to the employment and wage trends we observe

Future Work

- Add more occupation-level variables from BLS data

  - education requirements

  - projected job growth
  -
  - remote-work rates

- This would help determine whether AI exposure still predicts job growth after controlling for these factors